

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Fourth Semester of B. Tech. (IT) Examination

April 2013

IT207 Computer Networking - I

Date: 24.04.2013, Wednesday

Time: 01:30 p.m. To 04:30 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I

Q - 1 (a) Attempt the following questions.

[07]

- 1) Which layer is responsible for process to process message delivery?
- 2) For ring type of network topology, discuss the implication of single cable fault.
- 3) Which error detection method consists of just one redundant bit per data unit?
- 4) What is the mechanism of ENQ/ACK?
- 5) What is the function of router?
- 6) What is protocol?
- 7) Define Shannon capacity.

Q - 2 (a) Write a short note on Pulse Code Modulation technique.

[07]

(b) Why null modem is required? Explain in brief with diagram.

[05]

OR

(b) Explain time slot interchange method in switching.

[05]

(c) Assume a data stream 1011000001111011 encode this stream using NRZ-I encoding schemes.

[02]

OR

(c) Assume a data stream 1011000001111011 encode this stream using HDB3 encoding schemes.

[02]

Q - 3 (a) Explain the nature of light in optical fiber cable.

[07]

(b) Classify multiplexing techniques. Which technique is best suited for analog and digital. Why?

[03]

(c) Give the difference between circuit-switched network and packet-switched network.

[04]

OR

Q - 3 (a) What is guided media? Discuss twisted pair cable in detail.

[07]

(b) Draw the asynchronous TDM for following four signal information: (frame size=3)

[03]

source 1: T E G

source 2: A

source 3:

source 4: E F I L

(c) Explain FDDI in detail.

[04]

SECTION – II

Q - 4 (a) Define the following :

[07]

1. Baud rate
2. Periodic signal
3. Frequency
4. Amplitude
5. Encoding
6. Digital signal
7. Critical angle

Q - 5 (a) List out the functionality of transport layer. Briefly explain each one.

[07]

(b) Explain line configuration with example.

[04]

(c) List out the networking and internetworking devices with corresponding OSI layer name.

[03]

OR

Q - 5 (a) List out the functionality of presentation layer. Briefly explain each one.

[07]

(b) Write down the advantages and disadvantages of mesh topology.

[04]

(c) Define : physical address, logical address, port address

[03]

Q - 6 (a) Given 10-bit data sequence 1010011110 and a divisor of 1011, find the CRC.

[07]

(b) How does single bit error differ from burst error? Discuss the concept of redundancy in error detection and correction.

[04]

(c) Explain error control of stop and wait ARQ.

[03]

OR

Q - 6 (a) For data bit sequence 110110001010, find out the redundant bits using hamming code.

[07]

(b) Differentiate between error control and flow control. How they are useful in communication system?

[04]

(c) What does the number on a NAK frame mean for stop and wait, go-back-n and selective-reject ARQ?

[03]
